save

GameSaveService
- serializer: GameSerializer

+ save(game: Game, filePath: String): void
+ load(filePath: String): Game

uses 1 → 1

GameSerializer

+ serialize(game: Game): String
+ deserialize(data: String): Game

request

DrawTrainCardRequest
- slot: int

ClaimRouteRequest
- routeId: int
- selectedTrainCardIds: List<Integer>

KeepDrawDestRequest
- keepDestCardsIds: List<Integer>
- discardDestCardsIds: List<Integer>

DrawDestCardRequest

<<interface>> ActionRequest

+ buildCommand(game: Game): ActionCommand

<<interface>> GameUI

+ onTrainCardClicked(slot: int): void
+ onClaimRouteClicked(routeId: int, cardIds: List<Integer>): void
+ onDrawDestinationCardsClicked(): void
+ onTicketsSelectionConfirmed(keepIds: List<Integer>, discardIds: List<Integer>): void

bus 1 ↔ 1

contains 1 → 1

main

<<interface>> PendingDecisionVisitor

+ visitTicketDecision(decision: TicketDecision): void

Game
- players: List<Player>
- currentPlayerIndex: int
- board: Board
- trainCardDeck: TrainCardDeck
- destinationCardDeck: DestinationCardDeck
- endGameService: EndGameService
- routeScoringStrategy: RouteScoringStrategy
- turnMediator: TurnMediator
- lastRoundTracker: LastRoundTracker

+ play(request: ActionRequest): ActionResult
+ endTurn(): ActionResult
+ getCurrentPlayer(): Player
+ isGameOver(): boolean
+ resolveGameOver(): List<Player, List<Player>>
+ getTrainCardDeck(): TrainCardDeck
+ getDestinationCardDeck(): DestinationCardDeck
+ getPlayerByIndex(ind: int): Player

GameController
- game: Game
- gameSaveService: GameSaveService

+ onStartGame(config: GameConfiguration): void
+ handleRequest(actionReq: ActionRequest): ActionResult
+ notifyPendingDecision(pendingDecision: PendingDecision): void
- processActionResult(actionResult: ActionResult): void
- processEndTurnResult(endTurnResult: EndTurnResult): void
+ saveGame(filePath: String): void
+ loadGame(filePath: String): void

1 manages 1

contains 1 → 1

setup

PlayerConfiguration
- playerName: String
- playerColour: PlayerColour

+ getPlayerName(): String
+ getPlayerColour(): PlayerColour

includes 2 → 1

GameConfiguration
- playerConfigList: List<PlayerConfiguration>
- map: MapType

+ getPlayerConfigs(): List<PlayerConfiguration>
+ getMapType(): MapType

selects 1 → 1

<<enumeration>> MapType

LONDON

LondonGameBuilder
- newGame: Game

<<interface>> GameBuilder

+ buildBoard(): void
+ buildDecks(): void
+ buildPlayers(playerConfigs: List<PlayerConfiguration>): void
+ buildServices(): void
+ dealStartingHands(): void
+ getResult(): Game

GameSetupDirector

+ constructGame(config: GameConfiguration): Game

command

<<enumeration>> ActionCost

FREE
FULL
HALF
- cost: int

+ getCost(): int

<<interface>> ActionCommand

+ execute(player: Player): ActionResult
+ getCost(): ActionCost

ActionResult
- status: ActionStatus
- message: String
- endTurnOutcome: EndTurnOutcome
- pendingDecision: PendingDecision

+ isSuccess(): boolean
+ getEndTurnResult(): EndTurnResult
+ static success(message: String): ActionResult
+ static fail(message: String): ActionResult
+ static pending(message: String, decision: PendingDecision): ActionResult
+ setEndTurnResult(EndTurnResult): void
+ getPendingDecision(): PendingDecision
+ hasPendingDecision(): boolean

has 1 → 0..1

<<interface>> PendingDecision

+ accept(visitor: PendingDecisionVisitor): void

<<enumeration>> EndTurnOutcome

NEXT_PLAYER_TURN
LAST_ROUND_TRIGGERED
LAST_ROUND_CONTINUES
GAME_OVER

<<enumeration>> ActionStatus

SUCCESS
FAILED
PENDING
PENDING_FAILED

with status 1 → 0..1

TicketDecision
- drawnCards: List<DestinationCard>
- minKeep: int

+ getDrawnCards(): List<DestinationCard>
+ getMinKeep(): int
+ getCardsbyIds(ids: List<Integer>): List<DestinationCard>

ClaimRouteCommand
- route: Route
- selectedTrainCards: List<TrainCard>
- trainCardDeck: TrainCardDeck
- scoring: RouteScoringStrategy

DrawTrainCardCommand
- slot: int
- trainCardDeck: TrainCardDeck

InvalidCommand
- errorMessage: String
- isPending: boolean

KeepDestCommand
- keepDestCards: List<DestinationCard>
- discardDestCards: List<DestinationCard>
- ticketDeck: DestinationCardDeck
- minKeep: int

DrawDestCommand
- ticketDeck: DestinationCardDeck
- minKeep: int

updates 1 → 1

processes 1 → 1

draws 1 → 1

endgame

<<interface>> RouteScoringStrategy

+ score(route: Route): int

StandardRouteScoring
- scoreTable: Map<Integer, Integer> {readOnly}

EndGameService
- finalScores: Map<Player, Integer>
- endGameScoring: List<EndGameScoringStrategy>
- tieBreakerStrategy: List<TieBreakerStrategy>

+ endGameScoring(players: List<Player>, board: Board): void
+ determineWinner(players: List<Player>, board: Board): Player

retains 1 → 0..*

contains 1 → 0..*

<<interface>> EndGameScoringStrategy

+ apply(players: List<Player>, board: Board): void

<<interface>> TieBreakerStrategy

+ getTieBreaker(List<Player>, Board): List<Player>
+ getTieOrder(): int

DestinationTicketScoring
- playerDestCompletedCount: Map<Player, Integer>
- tieOrder: int {readOnly}

LongestRouteScoring
- BONUS: int = 10 {readOnly}
- bonusPlayer: Set<Player>
- tieOrder: int {readOnly}

- computeLongestPath(player: Player, board: Board): int

finalizes scores 1 → 1

refers 1 → 1

character

Player
- name: String
- colour: PlayerColour
- remainingTrainPieces: int
- score: int
- hand: PlayerHand
- ability: PlayerAbility

+ getHand(): PlayerHand
+ getScore(): int
+ getRemainingTrainPieces(): int
+ getName(): String
+ getColour(): PlayerColour
+ addScore(points: int): void
+ hasEnoughTrainPieces(count: int): boolean
+ reduceTrainPieces(numOfPieces: int): void
+ claimRoute(Route, List<TrainCard>, RouteScoringStrategy): void
+ drawTrainCard(TrainCard): void
+ drawDestCard(List<DestinationCard>): void

has 1 → 0..1

<<enumeration>> PlayerColour

RED
BLUE
GREEN
YELLOW
BLACK

PlayerHand
- trainCards: Map<CardColour, List<TrainCard>>
- destinationCards: List<DestinationCard>

+ getTrainCards(): Map<CardColour, List<TrainCard>>
+ getDestinationCards(): List<DestinationCard>
+ getLocomotiveCount(): int
+ getTrainCardCountByColour(colour: CardColour): int
+ addTrainCards(cards: List<TrainCard>): void
+ addDestinationCards(cards: List<DestinationCard>): void
+ removeTrainCards(cards: List<TrainCard>): void
+ removeDestinationCards(cards: List<DestinationCard>): void
+ hasEnoughOfAnyColour(needed: int): boolean
+ getTrainCardsByIds(ids: List<Integer>): List<TrainCard>
+ getDestinationCardByIds(ids: List<Integer>): List<TrainCard>

holds 1 → 0..*

holds 1 → 0..*

card

<<abstract>> Card
- id: int

+ getId(): int

TrainCard
- colour: CardColour {readOnly}

+ getColour(): CardColour
+ isLocomotive(): boolean

DestinationCard
- from: City
- to: City
- points: int

+ getFrom(): City
+ getTo(): City
+ getPoints(): int
+ isCompleted(ownerPlayer: Player, board: Board): boolean

<<enumeration>> CardColour

ORANGE
YELLOW
GREEN
BLUE
PINK
BLACK
WILD

uses 1 → 1

claims 1 → 0..*

holds 1 → 0..*

holds 1 → 0..*

deck

<<abstract>> Deck
- inDeckPile: List<Card>

+ draw(): Card
+ shuffle(): void
+ getInDeck(): List<Card>
+ isInDeckEmpty(): boolean

TrainCardDeck
- discardedPile: List<TrainCard>
- faceUpPile: List<TrainCard>

+ canDraw(slot: int): boolean
+ draw(slot: int): TrainCard
+ isFaceUpEmpty(): boolean
+ getFaceUpCards(): List<TrainCard>
+ discard(cards: List<TrainCard>): void
+ canDrawFaceUp(slot: int): boolean
+ peekFaceUp(slot: int): TrainCard
+ drawFaceUp(slot: int): TrainCard
- drawInDeck(): TrainCard
- checkAndReplaceLocomotives(): void
- shuffleDiscardIntoDeck(): void
- replaceFaceUp(slot: int): void
- tryRefillFaceUpPile(): void
- tryRefillInDeckPile(): void

DestinationCardDeck

+ drawThree(): List<DestinationCard>
+ returnToBottom(cards: List<DestinationCard>): void

holds 1 → 0..*

holds 1 → 0..*

tracks 1 → 1

turnmanager

TurnMediator
- remainingActionCost: int
- pendingDecision: PendingDecision
- MAX_ACTION_COST: int = 2 {readOnly}

+ isTurnComplete(): boolean
+ processAction(action: ActionCommand, currentPlayer: Player): ActionResult
- clearPendingDecision(): void
- reset(): void

LastRoundTracker
- triggered: boolean
- turnsRemaining: int
- triggeringPlayer: Player
- totalPlayers: int

+ endTurn(currentPlayer: Player): EndTurnOutcome
+ checkAndTrigger(currentPlayer: Player): boolean
+ decrementRemainingTurns(): void
+ isTriggered(): boolean
+ isOver(): boolean
+ getTriggeringPlayer(): Player

references 1 → 1

triggered by 1 → 1

ability

<<interface>> PlayerAbility

+ modifyRouteCost(): void
+ modifyRouteScore(): void
+ getDescription(): string

FerryMasterAbility

DistanceMasterAbility

ColourMasterAbility

has 1 → 1

map

Board
- cities: List<City>
- routes: List<Route>

+ getRoute(from: City, to: City): Route
+ getRouteById(routeID: string): Route
+ getRoutesClaimedBy(player: Player): List<Route>
+ isPathConnected(player: Player, from: City, to: City): boolean
+ getAvailableRoutes(): List<Route>

contains 1 → 1..*

City
- name: String
- x: int
- y: int

+ getName(): String
+ getX(): int
+ getY(): int

connects 1..* → 1..*

Route
- id: String
- cityA: City
- cityB: City
- length: int
- colour: RouteColour
- isClaimed: boolean
- owner: Player
- claimStrategy: RouteClaimStrategy
- routeGroup: RouteGroup

+ getId(): String
+ getCityA(): City
+ getCityB(): City
+ getLength(): int
+ getColour(): RouteColour
+ isClaimed(): boolean
+ setOwner(player: Player): void
+ getOwner(): Player
+ getClaimStrategy(): RouteClaimStrategy
+ canClaim(player: Player, trainCards: List<TrainCard>): boolean

RouteGroup
- routeInParallel: List<Route>
- playerCount: int {read only}

+ canClaim(player: Player, route: Route): boolean

contains 0..* → 1

has 1 → 1

<<enumeration>> RouteColour

ORANGE
YELLOW
GREEN
BLUE
PINK
BLACK
GREY

claimed by 1 → 1

claimed with 1 → 1

strategy

<<interface>> RouteClaimStrategy

+ validate(player: Player, route: Route, spentCards: List<TrainCard>): boolean

FerryRouteClaimStrategy

StandardRouteClaimStrategy

GrayRouteClaimStrategy